Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Which of the following is / are true?
I. If A is a subset of the universal set U,
then its complement A' is also a subset of U.
II. If $U = \{1, 2, 3, \ldots\ldots, 10\}$ and $\mathrm{A} = \{1, 3, 5, 7, 9\}$, then $(\mathrm{A}')' = \mathrm{A}$

(A) Only I is true

(B) Only II is true

(C) Both I and II are true

(D) None of these

Q.2

For a set A, consider the following statements:
1. $A \cup P(A) = P(A)$
2. $\{A\} \cap P(A) = A$
3. $P(A) - \{A\} = P(A)$

where P denotes power set. Which of the statements given above is/are correct?

(A) 1 only

(B) 2 only

(C) 3 only

(D) 12 and 3

Q.3

If A and B are two disjoint sets, then which one of the following is correct?

(A) $A - B = A - (A \cap B)$

(B) $B - A' = A \cap B$

(C) $A \cap B = (A - B) \cap B$

(D) All of these

Q.4

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Let N denote the set of natural numbers and $A = \{n^2 : n \in N\}$ and $B = \{n^3 : n \in N\}$. Which one of the following is incorrect?

(A) $A \cup B = N$

(B) The complement of $(A \cup B)$ is an infinite set

(C) $A \cap B$ must be a finite set

(D) $A \cap B$ must be a proper subset of $\{\mathrm{m}^6 : \mathrm{m} \in N\}$

Q.5

If the cardinality of a set A is 4 and that of a set B is 3 , then what is the cardinality of the set $A \Delta B$?

(A) 1

(B) 5

(C) 7

(D) Cannot be determined as the sets A and B are not given

Q.6

Let N be the set of non-negative integers, I the set of integers, $\mathrm{N_p}$ the set of non-positive integers, E the set of even integers and P the set of prime numbers. Then

(A) $\mathrm{I} - \mathrm{N} = \mathrm{N_p}$

(B) $\mathrm{N} \cap \mathrm{N_p} = \phi$

(C) $E \cap P = \phi$

(D) $\mathrm{N} \Delta \mathrm{N_p} = \mathrm{I} - \{0\}$

Q.7

$\mathrm{A}, \mathrm{B}, \mathrm{C}$ and D are four sets such that $\mathrm{A} \cap \mathrm{B} = \mathrm{C} \cap \mathrm{D} = \phi$. Consider the following :
1. $\mathrm{A} \cup \mathrm{C}$ and $\mathrm{B} \cup \mathrm{D}$ are always disjoint.
2. $\mathrm{A} \cap \mathrm{C}$ and $\mathrm{B} \cap \mathrm{D}$ are always disjoint Which of the above statements is/are correct ?

(A) 1 only

(B) 2 only

(C) Both 1 and 2

(D) Neither 1 nor 2 s

Q.8

Let $\mathrm{A}, \mathrm{B}, \mathrm{C}$ are three non-empty sets. If $\mathrm{A} \subset \mathrm{B}$ and $\mathrm{B} \subset \mathrm{C}$, then which of the following is true?

(A) $\mathrm{B} - \mathrm{A} = \mathrm{C} - \mathrm{B}$

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(B) $A \cap B \cap C = B$

(C) $A \cup B = B \cap C$

(D) $A \cup B \cup C = A$

Q.9

Let X and Y be two non-empty sets such that $X \cap A = Y \cap A = \phi$ and $X \cup A = Y \cup A$ for some non-empty set A. Then

(A) X is a proper subset of Y

(B) Y is a proper subset of X

(C) $X = Y$

(D) X and Y are disjoint sets

Q.10

Which one of the following is correct?
Here $\mathrm{P(A)}$ denotes the power set of a set A.

(A) $A \cup P(A) = P(A)$

(B) $A \cap P(A) = A$

(C) $A - P(A) = A$

(D) $\mathrm{P(A)} - \{\mathrm{A}\} = \mathrm{P(A)}$

Q.11

If $(\mathrm{A} - \mathrm{B}) \cup (\mathrm{B} - \mathrm{A}) = \mathrm{A}$ for subsets A and B of the universal set U, then which one of the following is correct?

(A) B is proper non-empty subset of A

(B) A and B are non-empty disjoint sets

(C) $\mathrm{B} = \phi$

(D) None of the above

Q.12

Let $n(\mathrm{U}) = 700, n(\mathrm{A}) = 200, n(\mathrm{B}) = 300, n(\mathrm{A} \cap \mathrm{B}) = 100$, then $n\left(\mathrm{A}' \cap \mathrm{B}'\right)$ is equal to

(A) 400

(B) 600

(C) 300

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(D) None of these

Q.13

There are 600 student in a school. If 400 of them can speak Telugu, 300 can speak Hindi, then the number of students who can speak both Telugu and Hindi is:

(A) 100

(B) 200

(C) 300

(D) 400

Q.14

In a group of 50 people, two tests were conducted, one for diabetes and one for blood pressure. 30 people were diagnosed with diabetes and 40 people were diagnosed with high blood pressure. what is the minimum number of people who were having diabetes and high blood pressure ?

(A) 0

(B) 10

(C) 20

(D) 30

Q.15

Out of 32 persons, 30 invest in National Savings Certificates and 17 invest in shares. What is the number of persons who invest in both?

(A) 13

(B) 15

(C) 17

(D) 19

Q.16

Which of the following is correct?
I. $\mathrm{n(S \cup T)}$ is maximum when $\mathrm{n(S \cap T)}$ is least.
II. If $\mathrm{n(U) = 1000, n(S) = 720, n(T) = 450}$, then least value of $n(S \cap T) = 170$

(A) Only I is true

(B) Only II is true

(C) Both I and II are true

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(D) Both I and II are false

Q.17

In a school, there are 20 teachers who teach Mathematics or Physics of these, 12 teach Mathematics and 4 teach both Maths and Physics. Then the number of teachers teaching only Physics are

(A) 4

(B) 8

(C) 12

(D) 16

Q.18

Given $n(\mathrm{U})=20, n(\mathrm{~A})=12, n(\mathrm{~B})=9, n(\mathrm{~A} \cap \mathrm{B})=4$
where U is the universal set, A and B are subsets of U, then $n\left((A \cup B)^{c}\right)=$

(A) 17

(B) 9

(C) 11

(D) 3

Q.19

In a town of 10000 families, it was found that 40% families buy newspaper A, 20% families buy newspaper B and 10% families buy newspaper $\mathrm{C}, 5\%$ buy A and $\mathrm{B}, 3\%$ buy B and C and 4% buy A and C. If 2% families buy all of three newspapers, then the number of families which buy A only, is

(A) 4400

(B) 3300

(C) 2000

(D) 500

Q.20

In a class of 30 pupils, 12 take needle work, 16 take physics and 18 take history. If all the 30 students take at least one subject and no one takes all three then the number of pupils taking 2 subjects is

(A) 16

(B) 6

(C) 8

(D) 20

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Q.1 Answer:

Both I and II are true

Solution:

If A is a subset of the universal set U, then its complement A' is also a subset of U. We have, $\mathrm{A}' = \{2, 4, 6, 8, 10\}$
Hence, $(\mathrm{A}')' = \{\mathrm{x} : \mathrm{x} \in \mathrm{U}$ and $\mathrm{x} \notin \mathrm{A}'\} = \{1, 3,$
$5, 7, 9\} = \mathrm{A}$
It is clear from the definition of the complement that for any subset of the universal set U, we have $(\mathrm{A}')' = \mathrm{A}$

Q.2 Answer:

1 only

Solution:

Since, Power set is the collection of all the subsets of the set A therefore $\mathrm{AUP}(\mathrm{A}) = \mathrm{P}(\mathrm{A})$
$\therefore$ statement (1) is correct.

Q.3 Answer:

$A - B = A - (A \cap B)$

Solution:

Since, A and B are two disjoints therefore $A \cap B = \phi$
$\therefore \quad \mathrm{A} - \mathrm{B} = \mathrm{A} - (\mathrm{A} \cap \mathrm{B})$

Q.4 Answer:

$A \cup B = N$

Solution:

Let $\mathrm{A} = \{n^2 : n \in \mathrm{N}\}$ and $\mathrm{B} = \{n^3 : n \in \mathrm{N}\}$
$\mathrm{A} = \{1, 4, 9, 16, \ldots..\}$
and $\mathrm{B} = \{1, 8, 27, 64, \ldots\ldots\}$
Now, $\mathrm{A} \cap \mathrm{B} = \{1\}$ which is a finite set.
Also, $A \cup B = \{1, 4, 8, 9, 27, \ldots..\}$
So, complement of $\mathrm{A} \cup \mathrm{B}$ is infinite set. Hence, $A \cup B \neq N$

Q.5 Answer:

Cannot be determined as the sets A and B are not given

Solution:

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Given, $n(A) = 4, n(B) = 3$

Since, the sets A and B are not known, then cardinality of the set $A \Delta B$ cannot be determined.

Q.6 Answer:

$\mathrm{N} \Delta \mathrm{N_p} = \mathrm{I} - \{0\}$

Solution:

$\mathrm{N} \Delta \mathrm{N_P} = (\mathrm{N} - \mathrm{N_p}) \cup (\mathrm{N_P} - \mathrm{N})$
$= \{1, 2, \ldots\} \cup \{\ldots -2, -1\} = I - \{0\}$

Q.7 Answer:

2 only

Solution:

Let $\mathrm{A} = \{1, 2\}$
$\mathrm{B} = \{3, 4, 0\}$
$\mathrm{C} = \{5, 6, 0\}$
$\mathrm{D} = \{7, 8\}$
Such that $(\mathrm{A} \cap \mathrm{B}) = (\mathrm{C} \cap \mathrm{D}) = \phi$
$\Rightarrow (\mathrm{A} \cup \mathrm{C}) = \{1, 2, 5, 6, 0\}$
$\Rightarrow (\mathrm{B} \cup \mathrm{D}) = \{3, 4, 7, 8, 0\}$
$\Rightarrow (\mathrm{A} \cup \mathrm{C}) \cap (\mathrm{B} \cup \mathrm{D}) = \{0\}$
So $(\mathrm{A} \cup \mathrm{C})$ and $(\mathrm{B} \cup \mathrm{D})$ are not always disjoint $\Rightarrow (\mathrm{A} \cap \mathrm{C}) = \phi$ and $(\mathrm{B} \cap \mathrm{D}) = \phi$
So $(A \cap C)$ and $(B \cap D)$ are always disjoint.

Q.8 Answer:

$A \cup B = B \cap C$

Solution:

If $A \subset B$ and $B \subset C$, then these sets 15 represented in Venn diagram as
Clearly, $\mathrm{A} \cup \mathrm{B} = \mathrm{B}$
and $\mathrm{B} \cap \mathrm{C} = \mathrm{B}$
Hence, $A \cup B = B \cap C$.

Q.9 Answer:

$X = Y$

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Solution:

Suppose $a \in X$ **and** $a \in A \Rightarrow a \in X \cup A \Rightarrow$
$a \in Y \cup A$
$\Rightarrow a \in Y$ **and** $a \in A(\because X \cup A = Y \cup A)$
$\Rightarrow a \in Y \cap A \Rightarrow Y \cap A$ **is non-empty**
This contradicts that $Y \cap A = \phi$ **So** $X = Y$

Q.10 Answer:

$A \cup P(A) = P(A)$

Solution:

$\mathrm{A} \cup \mathrm{P}(\mathrm{A}) = \mathrm{P}(\mathrm{A})$ **is correct.**
Since A **is a subset of its power set**

Q.11 Answer:

$\mathrm{B} = \phi$

Solution:

For subsets A **and** B **of** U, $\mathrm{If}(\mathrm{A} - \mathrm{B}) \cup (\mathrm{B} - \mathrm{A}) = \mathrm{A}$
$\Rightarrow \mathrm{B} = \phi$

Q.12 Answer:

300

Solution:

$n\left(\mathrm{A}' \cap \mathrm{B}'\right) = n(\mathrm{A} \cup \mathrm{B})' = n(\mathrm{U}) - n(\mathrm{A} \cup \mathrm{B})$
$= n(\mathrm{U}) - [n(\mathrm{A}) + n(\mathrm{B}) - n(\mathrm{A} \cap \mathrm{B})]$
$= 700 - [200 + 300 - 100] = 300$

Q.13 Answer:

100

Solution:

Let $\mathrm{A} \equiv$ **Set of Tamil speaking students and** $\mathrm{B} \equiv$ **Hindi speaking students** $n(A) = 400, n(B) = 300$ **and** $n(A \cup B) = 600$
$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
$\Rightarrow \mathrm{n}(\mathrm{A} \cap \mathrm{B}) = \mathrm{n}(\mathrm{A}) + \mathrm{n}(\mathrm{B}) - \mathrm{n}(\mathrm{A} \cup \mathrm{B})$
$= 400 + 300 - 600 = 100$

Q.14 Answer:

20

Solution:

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$n(T) = 50$
$n(D) = 30$
$n(H) = 40$
$n(T) = n(D) + n(H) - n(D \cap H)$
$50 = 30 + 40 - n(D \cap H)$
$n(D \cap H) = 70 - 50 = 20$
Number of people having diabetes and high blood pressure $= 20$

Q.15 Answer:

15

Solution:

Let N **= National savings certificates**
S **= Shares**
Total no. of persons $= 32$
No. of persons who invest in National savings certificates $= 30$
No. of persons who invest in shares $= 17$ **Therefore**
$n(N \cup S) = 32, n(N) = 30, n(S) = 17$
We know that,
$n(N \cup S) = n(N) + n(S) - n(N \cap S)$
$\Rightarrow \quad 32 = 30 + 17 - n(N \cap S)$
$\Rightarrow n(N \cap S) = 47 - 32 = 15$

Q.16 Answer:

Both I and II are true

Solution:

Both the statements are true.
II. $n(S \cup T) = n(S) + n(T) - n(S \cap T)$
$= 720 + 450 - n(S \cap T)$
$= 1170 - n(S \cap T)$
$\Rightarrow \quad 1170 - n(S \cap T) \leq n(U) \Rightarrow 1170 - n(S \cap T)$
≤ 1000
$\Rightarrow \quad n(S \cap T) \geq 170$

Q.17 Answer:

8

Solution:

Let M **= set of Mathematics teachers**
P **= set of Physics teachers**

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$\mathrm{n}($ **only Maths teacher** $) = n(M) - n(M \cap P) = 12 - 4 = 8$
Also, $\mathrm{n}(\mathrm{M} \cup \mathrm{P}) = \mathrm{n}$ **(only Math teachers)** $+\mathrm{n}($ **Only Physics teachers** $) + \mathrm{n}(\mathrm{M} \cap \mathrm{P})$
$20 = 8 + 4 + \mathrm{n}($ **only Physics teachers** $) \Rightarrow \mathrm{n} = 8$

Q.18 Answer:

3

Solution:

$n(\mathrm{A} \cup \mathrm{B}) = n(\mathrm{A}) + n(\mathrm{B}) - n(\mathrm{A} \cap \mathrm{B})$
$= 12 + 9 - 4 = 17$
Now, $n\left((\mathrm{A} \cup \mathrm{B})^{\mathrm{C}}\right) = n(\mathrm{U}) - n(\mathrm{A} \cup \mathrm{B})$
$= 20 - 17 = 3$

Q.19 Answer:

3300

Solution:

$\mathrm{n}(\mathrm{A}) = 40\%$ **of** $10000 = 4000, \mathrm{n}(\mathrm{B}) = 2000$
$\mathrm{n}(\mathrm{C}) = 1000, \mathrm{n}(\mathrm{A} \cap \mathrm{B}) = 500, \mathrm{n}(\mathrm{B} \cap \mathrm{C}) = 300$
$n(C \cap A) = 400, n(A \cap B \cap C) = 200$
$\mathrm{n}(\mathrm{A} \cap \overline{\mathrm{B}} \cap \overline{\mathrm{C}}) = \mathrm{n}\{\mathrm{A} \cap (\mathrm{B} \cup \mathrm{C})'\}$
$= \mathrm{n}(\mathrm{A}) - \mathrm{n}\{\mathrm{A} \cap (\mathrm{B} \cup \mathrm{C})\}$
$= \mathrm{n}(\mathrm{A}) - \mathrm{n}(\mathrm{A} \cap \mathrm{B}) - \mathrm{n}(\mathrm{A} \cap \mathrm{C}) + \mathrm{n}(\mathrm{A} \cap \mathrm{B} \cap \mathrm{C})$
$= 4000 - 500 - 400 + 200 = 3300$

Q.20 Answer:

16

Solution:

Given $n(\mathrm{N}) = 12, n(\mathrm{P}) = 16, n(\mathrm{H}) = 18, n(\mathrm{N} \cup$
$\mathrm{P} \cup \mathrm{H} = 30$
From, $n(\mathrm{N} \cup \mathrm{P} \cup \mathrm{H}) = n(\mathrm{N}) + n(\mathrm{P}) + n(H) - n(\mathrm{N} \cap \mathrm{P})$
$-n(\mathrm{P} \cap \mathrm{H}) - n(\mathrm{N} \cap \mathrm{H}) + n(\mathrm{N} \cap \mathrm{P} \cap \mathrm{H})$
$\therefore \quad n(\mathrm{N} \cap \mathrm{P}) + n(\mathrm{P} \cap \mathrm{H}) + n(\mathrm{N} \cap \mathrm{H}) = 16$
Now, number of pupils taking two subjects $= n(\mathrm{N} \cap \mathrm{P}) + n(\mathrm{P} \cap \mathrm{H}) + n(\mathrm{N} \cap \mathrm{H}) -$
$3n(\mathrm{N} \cap \mathrm{P} \cap \mathrm{H})$
$= 16 - 0 = 16$

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